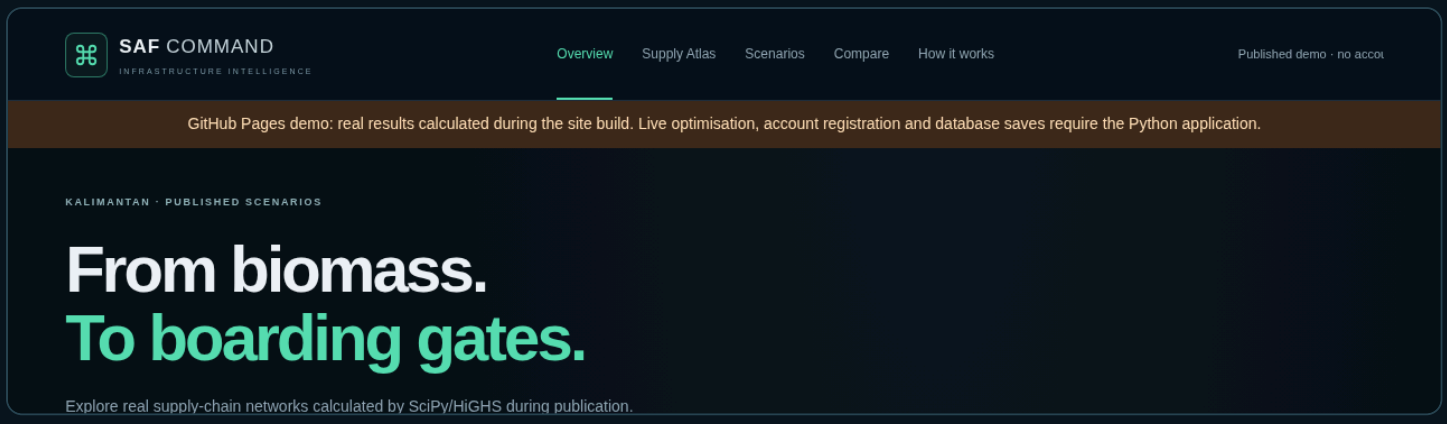
HACKATHON PROJECT BRIEF · 31 OCTOBER 2026

SAF COMMAND

From biomass to boarding gates. Every decision optimised.



80
biomass locations

6,800 kt
available biomass

4
airport destinations

The planning problem

Dispersed biomass, facility investment and airport supply must be coordinated. SAF Command jointly selects depot/refinery sites, sizes and three-stage material flows across Kalimantan.

The implementation

Next.js and MapLibre interface; FastAPI; SciPy/HiGHS MILP; PostgreSQL, SQLAlchemy and Alembic; Docker and GitHub Actions. Constraints enforce source supply, capacity, mass yield and exact airport deliveries.

Inspectable evidence

Costs, capacities, flows, input snapshots, provenance and actual solver gaps. Independent checks cover the mathematical constraints, accounts, ownership and exports.

What judges can use now

The free public site provides the atlas, three genuine build-time solves, comparisons, explanations and JSON/CSV downloads. No signup or AI key is required.

What requires Docker

Editable assumptions, live Python optimisation, accounts and database-backed scenario history are implemented in the full application. They are not hosted on GitHub Pages.

Model limits

Demands/yield are illustrative. Routes are straight-line proxies; sites need permitting/engineering checks. Unknown operating economics and lifecycle emissions remain outside the model.

Demo: https://w0mbler.github.io/SAF-Anthropic-Hackathon/
Code: https://github.com/W0mbler/SAF-Anthropic-Hackathon

MIT applies to new source only; supplied data rights require separate verification. No Claude/Anthropic integration is claimed. Development used OpenAI Codex assistance. Screenshots show the locally served Pages export; numerical results come from real solver runs.